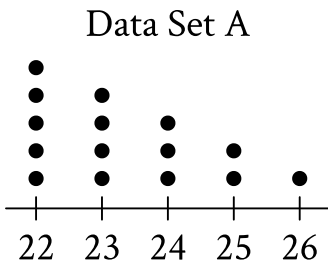


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Medium

Question



Data set A has **15** values and is represented by the dot plot shown. Data set B is created by adding **46** to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

Answer

- A. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
- B. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.
- C. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
- D. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

Correct Answer: A

Rationale

Choice A is correct. The median is the middle value in a data set when the data are arranged in order from least to greatest. Since there are **15** values in data set A, the median is the **8th** value. The **8th** value is **23**, so the median of data set A is **23**. The range is found by subtracting the minimum value in a data set from the maximum value. The minimum value in data set A is **22** and the maximum value is **26**. Therefore, the range of data set A is **26 — 22**, or **4**. It's given that data set B is created by adding **46** to each of the values in data set A. Therefore, the **8th** value in data set B is **23 + 46**, or **69**, so the median of data set B is **69**. The minimum value in data set B is **22 + 46**, or **68**, and the maximum value is **26 + 46**, or **72**. Therefore, the range of data set B is **72 — 68**, or **4**. Since the median of data set A is **23**, and the median of data set B is **69**, the median of data set B is greater than the median of data set A. Since the ranges of data sets A and B are both **4**, the range of data set B is equal to the range of data set A.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.